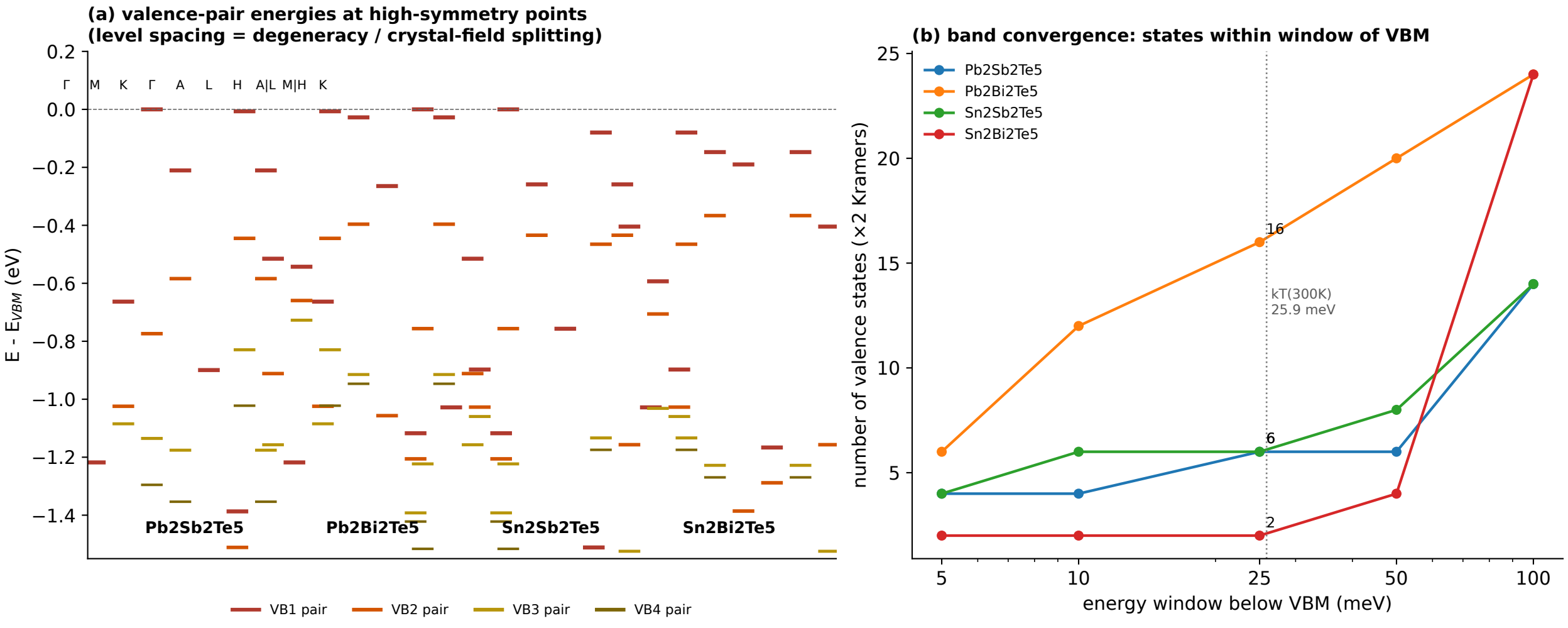
(a) valence-pair energies at high-symmetry points
(level spacing = degeneracy / crystal-field splitting)
Γ M K Γ A L H A|L M|H K
E - E_VBM (eV)
Pb2Sb2Te5
Pb2Bi2Te5
Sn2Sb2Te5
Sn2Bi2Te5
VB1 pair
VB2 pair
VB3 pair
VB4 pair
(b) band convergence: states within window of VBM
number of valence states (×2 Kramers)
energy window below VBM (meV)
kT(300K)
25.9 meV
16
6
2